Tech Mock Interview

Olga Sycheva | Software Engineering | Career Sprint

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FULL TRANSCRIPT

AI Interviewer · 06:56 PM Hungary Time · 12:56 PM ET

Hi! I'm your AI interviewer for the Tech Mock — Software Engineering. This is a required interview for your Career Sprint, and I'll ask you a set of questions one at a time. Take your time. Are you ready to begin?

Olga Sycheva · 06:56 PM Hungary Time · 12:56 PM ET

Yes.

AI Interviewer · 06:56 PM Hungary Time · 12:56 PM ET

Great. Let's get started.

Can you explain the difference between var, let, and const, and when you'd use each?

Olga Sycheva · 06:57 PM Hungary Time · 12:57 PM ET

Can you repeat the question, please?

AI Interviewer · 06:57 PM Hungary Time · 12:57 PM ET

Of course. Can you explain the difference between var, let, and const, and when you'd use each?

Olga Sycheva · 06:57 PM Hungary Time · 12:57 PM ET

So var, let, and const are used to declare variables in JavaScript. Var, uh, is the older way and has function scope. Let has block scope and always asks to change the value. Const also has block scope, but we cannot reassign it. I usually use const by default, let when the value needs to change, and I avoid var in modern JavaScript.

AI Interviewer · 06:57 PM Hungary Time · 12:57 PM ET

Thank you. You mentioned that you avoid var in modern JavaScript. Can you walk me through a specific situation where using var caused a problem, or where switching to let or const made a difference in your code?

Olga Sycheva · 06:58 PM Hungary Time · 12:58 PM ET

I haven't had a specific problem with var in my projects, but I understand the risk. For example, var can be assessed outside a block, which may cause an experienced behavior. Using let or const make the code easier to understand and helps prevent mistakes. That's why I prefer them in my projects.

AI Interviewer · 06:58 PM Hungary Time · 12:58 PM ET

Got it. Now, can you explain the MERN stack architecture and how the components — MongoDB, Express, React, and Node — interact with each other?

Olga Sycheva · 06:59 PM Hungary Time · 12:59 PM ET

The MERN stacks include MongoB- MongoDB, Express, React, and Node. React is a front end where users interact with the application. Node and Express handle the back end, including API request and business logic. MongoDB stores the application data. When the user pla- performs an action in React, the front end send a request to the back end. Express process the request, communicate it with MongoDB if needed, andsend a response back to React. So together, these technologies allow us to build a complete full-stack application.

AI Interviewer · 06:59 PM Hungary Time · 12:59 PM ET

Thank you. You described the flow from React to Express to MongoDB and back. Can you tell me about a specific project where you built something using this stack, and what your role was in putting those pieces together?

Olga Sycheva · 06:59 PM Hungary Time · 12:59 PM ET

Absolutely. So...

AI Interviewer · 06:59 PM Hungary Time · 12:59 PM ET

Take your time.

Olga Sycheva · 07:00 PM Hungary Time · 01:00 PM ET

Well, in, uh, my Workspace Hub project, I work on task comments f- future using the MERN stack. I connected the React front end to the Express API, which handled request and stored data in MongoDB. My main responsibility was implementing the comments feature, adding permission checks, and testing different user roles. I also made sure that only authorized users could edit or delete comments. This project helped me understand how allparts of full stack application work together.

AI Interviewer · 07:00 PM Hungary Time · 01:00 PM ET

Good. Shifting gears a little — how do you hand off your code or documentation for others to continue building on it?

Olga Sycheva · 07:00 PM Hungary Time · 01:00 PM ET

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AI Interviewer · 07:00 PM Hungary Time · 01:00 PM ET

Take your moment. Would you like me to repeat the question?

Olga Sycheva · 07:01 PM Hungary Time · 01:01 PM ET

So, I use GitHub to organize and share my code. I write clear README files explaining the project, how to install and run it, and the main features. I also use meaningful comment, commit messages, and add comments when the code is complex. My goal is to make it easier for another developer to understand my work and continue building on it.

AI Interviewer · 07:01 PM Hungary Time · 01:01 PM ET

You mentioned writing comments when the code is complex. Can you walk me through a specific time you handed off code to someone else, and how they found it to work with?

Olga Sycheva · 07:02 PM Hungary Time · 01:02 PM ET

Yes, absolutely. Uh, during my studying on Triple Ten program, um, I used to, um... I used to send some comments, but I haven't done a formal code hand- handoff to another developer. But I've shared my projects with Triple Ten reviewers through the GitHub with the comments. For example, when I submitted my portfolio, a reviewer tested it and found problems with the deployment and the missing fav icon.This feedback helped me understand what another person might experience when working with my project. It helped me to check, uh, accessibility to improve clear documentation before sharing my code.

AI Interviewer · 07:02 PM Hungary Time · 01:02 PM ET

Thank you for sharing that. Now, imagine you've identified a critical bug in a system that needs immediate attention. How would you communicate this issue to both your technical lead and a non-technical stakeholder, ensuring they both understand the urgency and impact?

Olga Sycheva · 07:03 PM Hungary Time · 01:03 PM ET

Well, first, I would inform my technical lead about the bug, explain what is causing it, if I know, and share how it affect the system. Uh, I would also explain the urgency and suggest possible solution. For a non-technical stakeholder, I would use simple language and focus on business impact, such as users being unable to access the application. Uh, I would explain what we are doing to fix and, uh, keep both updated.On our progress.

AI Interviewer · 07:03 PM Hungary Time · 01:03 PM ET

You mentioned focusing on business impact for the non-technical stakeholder. Can you walk me through a specific time you had to communicate something urgent or technical to someone without a technical background, and how that conversation went?

Olga Sycheva · 07:03 PM Hungary Time · 01:03 PM ET

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AI Interviewer · 07:03 PM Hungary Time · 01:03 PM ET

Take your time. Would you like me to repeat the question?

Olga Sycheva · 07:03 PM Hungary Time · 01:03 PM ET

Yes.

AI Interviewer · 07:03 PM Hungary Time · 01:03 PM ET

Can you walk me through a specific time you had to communicate something urgent or technical to someone without a technical background, and how that conversation went?

Olga Sycheva · 07:04 PM Hungary Time · 01:04 PM ET

Well, I haven't had that exact situation in software development role yet, but I have related example from my portfolio project. The website worked locally, but visitors could, uh, couldn't access it because of the Vercel settings. I learned to explain the problem in simple terms, focusing on how it affect users rather than technical details.

AI Interviewer · 07:04 PM Hungary Time · 01:04 PM ET

Got...

Olga Sycheva · 07:04 PM Hungary Time · 01:04 PM ET

I would use the same approach with non-technical stakeholders and clearly explain the next steps.

AI Interviewer · 07:04 PM Hungary Time · 01:04 PM ET

Got it. Let's turn to this scenario: you're working on a feature, and halfway through, a new requirement comes in that significantly changes the scope and introduces a tight deadline. How do you approach this situation, and what steps would you take to manage the change and ensure a successful outcome?

Olga Sycheva · 07:05 PM Hungary Time · 01:05 PM ET

First, I would clarify the new requirements and understand how they affect the deadline. I think this is important. Then I would review the exist- the existing work and identify what needs to change. I would prioritize the most important features and discuss the timeline with my team. If necessary, I would suggest moving less important tasks to a later release. I would also communicate progress regularly and test the changes carefully.

AI Interviewer · 07:05 PM Hungary Time · 01:05 PM ET

You mentioned discussing the timeline with your team and possibly moving tasks to a later release. Can you tell me about a specific time you had to reprioritize or adjust your work because something unexpected came up, and what you actually did?

Olga Sycheva · 07:05 PM Hungary Time · 01:05 PM ET

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AI Interviewer · 07:05 PM Hungary Time · 01:05 PM ET

Take your time. Are you still there?

Olga Sycheva · 07:06 PM Hungary Time · 01:06 PM ET

Yes. During my portfolio project, I received unexpected feedback about two issues. The website wasn't publicly acceptable and file icon was missing. I had to adjust my priorities, so I focused on deployment problem first because the visitors couldn't open my portfolio. Um, I realized that later. I checked the Vercel settings before moving on to smaller issue. This experience taught me to stay flexible and focus on what matters most to users.

AI Interviewer · 07:06 PM Hungary Time · 01:06 PM ET

Thank you. Now, how do you test your applications? What tools or frameworks have you used?

Olga Sycheva · 07:07 PM Hungary Time · 01:07 PM ET

Well, I used, uh, some tools and framework in my work, and usually, uh, it h- it helps, um, with the Chrome dev- dev tools and IPA testing. I checked, um, how features work in the browser, looks, look for console errors, and inspect network request and IPA responses. For example, in Workspace Hub, I tested different user permission, and in Flashcard, I checked API data tofind and fix errors. I haven't used automatics testing frameworks, uh, extensively yet, but I'm interesting in learning Jest and React Testing Library.

AI Interviewer · 07:07 PM Hungary Time · 01:07 PM ET

You mentioned testing different user permissions in Workspace Hub. Can you walk me through what that testing actually looked like — what you checked, and how you caught or fixed any issues?

Olga Sycheva · 07:08 PM Hungary Time · 01:08 PM ET

In Workspace Hub, I tested which users could edit or delete comments. I checked the permissions for comment authors, administrators, and regular users. I also tested direct API requests to make sure users couldn't bypass the front-end re- restriction. I found the front-end checks alone weren't enough, so I added server-side permission checks and tested the different roles again to confirm the rules worked correctly.

AI Interviewer · 07:08 PM Hungary Time · 01:08 PM ET

Good. Now, how do you stay current with new frameworks, tools, and libraries?

Olga Sycheva · 07:08 PM Hungary Time · 01:08 PM ET

Reading, exploring, finding solution, reading internet. I stay, uh, I stay current by following official documentation in TripleTen program, taking courses, and practicing with real projects. Um, so I'm learning React, Node, and TypeScript. Um, I also use GitHub developer commun- communities and AI tools to explore new technologies. I believe the best way to learn is to apply new knowledge in practice.

AI Interviewer · 07:08 PM Hungary Time · 01:08 PM ET

You mentioned using AI tools to explore new technologies. Can you give me a specific example of a time you used an AI tool to learn something new, and what you actually built or applied from that?

Olga Sycheva · 07:09 PM Hungary Time · 01:09 PM ET

Um, yes. I used AI while learning Node and TypeScript for my MeshAI project. Uh, at first, I had trouble understanding how to set up to backend and run the application, so I used AI to explain the project structure, terminal commands, and error message step by step. Then I applied what I learned in VS Code and tested the application myself. It helped me become more confident with the backend development.

AI Interviewer · 07:09 PM Hungary Time · 01:09 PM ET

Thank you. Now, tell me about a time you had to deliver difficult news or feedback to a colleague or manager. How did you approach the situation, and what was the outcome?

Olga Sycheva · 07:09 PM Hungary Time · 01:09 PM ET

Could you repeat the question, please?

AI Interviewer · 07:09 PM Hungary Time · 01:09 PM ET

Of course. Tell me about a time you had to deliver difficult news or feedback to a colleague or manager. How did you approach the situation, and what was the outcome?

Olga Sycheva · 07:10 PM Hungary Time · 01:10 PM ET

Well, I think during my workspace hub project, I had to explain to my teammates that our comments permission checks might not be secure enough. I explained the users could potentially bypass the front-end checks through direct API request. I focused on problem, not blaming anyone, suggesting adding service side validation. We agreed on approach and improved the permission checks. It taught me that difficult feedback should be clear, respectful, and focused on solution.

AI Interviewer · 07:10 PM Hungary Time · 01:10 PM ET

You mentioned your teammates agreed on the approach. How did they initially react when you raised the security concern, and was there any pushback?

Olga Sycheva · 07:11 PM Hungary Time · 01:11 PM ET

Well, um, it's easy to communicate when you understand. And so my teammates were open to discuss the concern, and there wasn't any major pushback. I explained how direct API requests could bypass front-end permission checks. We discussed the risk, and then we agreed that service-side validation was necessary. So then I focused on implementing and testing those checks. Overall, it was a constructivedecision.

AI Interviewer · 07:11 PM Hungary Time · 01:11 PM ET

Good. Now, thinking about a different kind of challenge — you are building a simple chat feature for a web app. What are the main pieces you would need, front end, back end, and where messages get stored, and how would they talk to each other?

Olga Sycheva · 07:12 PM Hungary Time · 01:12 PM ET

I would use React for the front end, where users can type, send, and view messages. On the back end, I would use Node and Express to handle requests and manage the message. MongoDB could store the messages, including the sender, text, and timestamp. When a user send a message, React send it to the back end, which saves it in MongoDB. Uh, then the message ap- appears in the chat. For real-time updates, I could use websockets to u- so users receive new messages without refreshing the page.

AI Interviewer · 07:12 PM Hungary Time · 01:12 PM ET

You mentioned WebSockets for real-time updates. Can you walk me through how that connection would actually work — what happens on both the front end and the back end when a new message is sent?

Olga Sycheva · 07:13 PM Hungary Time · 01:13 PM ET

First, the React front end opens a WebSocket connection to the server. The Node back end keeps the connection open. When a user send a message, the front end send it to the server through the WebSocket. The server checks the message, saves it in MongoDB, and send it to users in the chat. On the front end, React receive the new message and up- updates the chat immediately without refreshing the page.

AI Interviewer · 07:13 PM Hungary Time · 01:13 PM ET

Thank you. Last one — a user updates their profile and, for a moment, still sees their old details on another page. What could cause that, and how would you go about investigating it?

Olga Sycheva · 07:14 PM Hungary Time · 01:14 PM ET

I think, uh, that could happen 'cause of caching, delaying database updates, or outdated data in React state. First, I guess I would check the... if the new profile information was saved correctly in a database. Then I would use Chrome devel- DevTo- DevTools to inspect the API response and see if it returns the updated data. If the API returns the correct information, I would check the front-end state and caching. I would also test the issueagain to make sure the updated profile appears correctly on every page.

AI Interviewer · 07:14 PM Hungary Time · 01:14 PM ET

You mentioned checking the front-end state as one of your steps. Can you be more specific about what you would actually look for in the React state, and how you would go about fixing it if that turned out to be the cause?

Olga Sycheva · 07:14 PM Hungary Time · 01:14 PM ET

I would check if the React state still connects to the old profile information. For example, one page might have updated the data, but another page could still be using an old copy. So, yeah, that could be a problem. I would use React developer tools to inspect the state and check w- whether it updates after the API request. If the state is outdated, I would update the shared state or refresh the user data after saving the profile. ThenI would test both pages to make sure they display the correct information.

AI Interviewer · 07:15 PM Hungary Time · 01:15 PM ET

Thank you. That brings us to the end of the interview. I really appreciate the time and thought you put into your answers today. The interview is now complete and will be scored. Best of luck with the rest of your programme!